

CHU ANH DUC

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EDUCATION AND TRAINING

Hanoi University of Science and Technology

2023 - Present

BSc. in Data Science and Artificial Intelligence - Elite Technology Program

Hanoi, Vietnam

- **Cumulative GPA: 3.97/4.0 (Top 3/120 in the Faculty)**
- **Core courses:** Artificial Intelligence (A), Data Science (A), Machine Learning (A), Deep Learning (A), Algebra (A), Calculus (A), Probability and Statistics (A), Applied Statistics and Experimental Design (A)
- **Highlight achievements:**
 - Second Prize, Vietnam Mathematical Olympiad (2022, 2023) - top 20/500 contestants
 - First Prize in Algebra; Third Prize in Calculus, National Mathematical Olympiad (2024)

PUBLICATIONS

[1] **Chu, A. D.***, Pham, T. K.*, Vu, H. M.*, Nguyen, D. T.*, Nguyen, T. T., Truong, T. N., Son, M. H., Nguyen, T. T., & Nguyen, P. L. (2026). PADM: A Physics-aware Diffusion Model for Attenuation Correction. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. <https://doi.org/10.48550/arxiv.2511.06948>

[2] **Chu, A. D.***, Tran, T. K.*, Pham, Q. H., Nguyen, P. L., & Hoang, T. N. (2026). Cross-Modal Knowledge Distillation without Paired Data: Theoretical Foundations and Algorithms. *International Conference on Machine Learning (ICML)*. <https://icml.cc/virtual/2026/poster/60546>

*Equal contribution

RESEARCH EXPERIENCE

Undergraduate Researcher - AI4LIFE Lab, HUST

Hanoi, 2024 – Present

Project 1: Physics-Aware Diffusion Models for Medical Imaging

- Investigated SPECT acquisition physics to develop a CT-free attenuation correction method for Myocardial Perfusion Imaging.
- Implemented diffusion/GAN-based baselines for NAC-to-AC image translation on 73,680 cardiac SPECT slices, achieving 13.83% RMSE improvement over SOTA.
- **Skills:** Medical image processing pipelines, diffusion models, experimental benchmarking.

Project 2: Optimal Transport for Cross-Modal Knowledge Distillation

- Co-designed the first cross-modal knowledge distillation framework for unpaired data (audio and visual data).
- Aligned feature and prediction distributions across modalities using bilevel optimization with Wasserstein optimal transport.
- Implemented the full training pipeline and validated the theoretical bounds; tuned experiments across 4 benchmarks with VGGSound and 200,000+ videos using models (ViT) with up to 300 million parameters.
- Achieved +14.3% over Cross Entropy, +7.5% over Feature Distillation on all 8 unpaired tasks; ranked 1st on 4/6 paired tasks.
- **Skills:** optimal transport theory, bilevel optimization, and end-to-end experimental design.

Project 3: Theory of Long-Tailed Generation in Diffusion Models

- Deriving theoretical lower bounds on tail-class generation error for conditional DDPMs, characterizing regimes where classifier-free guidance provably fails on rare classes.
- **Skills:** score-based generative modeling, sample complexity analysis

ACADEMIC EXPERIENCE

Reviewer, Workshop Decision-Making from Offline Datasets to Online Adaptation, ICML 2026

- Reviewed 3 submissions for DEMO at ICML 2026, hosted by Prof. Jana Doppa (Washington State University).
- Explored safe and scalable generative models, causal ML, and AI for Science, focusing on uncertainty, benchmarks, and real-world evaluation.

Invited Participant, KAIST School of Computing Global Preview Week | Daejeon, South Korea - Sep 2026

- Selected for a fully funded one-week research visit at KAIST School of Computing; selected 20 from 1000+ participants.

SKILLS AND LANGUAGES

- **Programming:** Python, C++, Java, LaTeX
- **ML/DL Frameworks:** PyTorch, torchvision, timm, NumPy, SciPy, Pandas, Matplotlib, Weights & Biases
- **MLOps & Infrastructure:** Git, Linux/Bash, SSH, rsync, rclone, HuggingFace Hub
- **Research:** Optimal Transport, Knowledge Distillation, Diffusion Models, Bilevel Optimization, Flow Matching

Languages: English: Advanced level (IELTS 7.0)

EXTRACURRICULAR ACTIVITIES

Co-Lead, FPT Young Talent Global Webinar Program | FPT Young Talent (FYT) Hanoi, 2025 – Present

A highly selective technology talent community under FPT Corporation (top 15/200+ applicants)

- Organized an international webinar series, sharing career insights about AI, Finance, and entrepreneurship.
- Led a team of 8 to design 3 sessions, including DSAI, Finance, and Entrepreneurship.

Teaching Instructor | Academic Support Club, HUST Hanoi, 2024 – Present

A student club dedicated to providing academic support for undergraduates

- Delivered lectures on Linear Algebra to 60+ students, provided Q&A support for 200+ students; organized mock exams for 600+ on-site participants for midterm and final-term preparation.
- Provided revision materials in Linear Algebra, accessed by 2000+ undergraduates.

SCHOLARSHIPS AND AWARDS

Academic Excellence Scholarship (2024-2025) | Top 5% of the University

Awarded by Hanoi University of Science and Technology

Future VPBanker Scholarship (2025) | Top 54/600+ applicants

Scholarship for leadership and technology-finance potential, awarded by VPBank

Vallet Scholarship (2022, 2026) | Top 5/460 applicants

Excellence scholarship for outstanding students in STEM, awarded by Odon Vallet's Foundation

Gold Level, WorldQuant Brain Challenge (2025)

Global quantitative finance competition hosted by WorldQuant.

First Prize in National Student Startup Competition (2026) | Top 1/15 teams

Awarded by the Ministry of Education and Training.

REFERENCES

- **Assoc. Prof. Nguyen Phi Le**
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- **Assoc. Prof. Hoang Trong Nghia**
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